



Marine Conservation - OCEASTIC Project



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1. Existing Problem

Are floating barriers *the* solution to plastic pollution in Egypt? Plastic pollution stands as one of the most pressing environmental concerns of the 21st century, profoundly impacting ecosystems, human health, and economies worldwide. Annually, approximately 350–380 million tons of new plastic are produced (Shen & Worell, 2024), with only 9% being recycled, while the rest ends up as waste (Abdullah & Abedin, 2024).

Recent estimations conclude that 0,5% of the plastic waste ends up in the ocean or at coastlines annually (Ritchie, 2023). Ocean-based sources are responsible for 20% of plastic debris in marine environments, with commercial fishing being the primary human activity contributing to this (Li et al., 2016).

Once the plastic enters the ocean, it poses a severe and growing threat to environmental systems, socio-economic stability, and human health. Plastic pollution has already had harmful effects on the majority of marine species, with evidence showing negative impacts on 88% of them. Additionally, it is estimated that 90% of seabirds and over half of all sea turtles have ingested plastic debris (World Economic Forum, 2022). As a result, marine animals often suffer serious injuries, health problems, or painful deaths due to their interactions with plastic debris (Li et al., 2016).

In addition, microplastics, which are tiny plastic particles resulting from the breakdown of larger debris, disrupt plant photosynthesis and threaten food supply (Carrington, 2025). Consequently, nutrient cycles are disrupted, ultimately reducing biodiversity and harming sensitive ecosystems like coral reefs (Bubu-Davies & Anwuri, 2022; De-la-Torre, 2020).

Coastal communities and industries suffer significant economic losses due to plastic pollution. Tourism declines as polluted beaches deter visitors (Beaumont et al., 2019). Moreover, plastic pollution in the ocean reduces the productivity of commercial fisheries by harming fish health, disrupting marine ecosystems, and contributing to declining fish population (Wootton et al., 2022). Plastic pollution is, therefore, leading to significant economic losses (Beaumont et al., 2019). In the Mediterranean region alone, these sectors suffer combined losses estimated at €641 million annually due to plastic waste (Dalberg Advisors et al., 2019, p. 12). Furthermore, plastics that degrade into microplastics accumulate in for example seafood consumed by humans. These particles can carry harmful chemicals, posing significant risks to human health (Smith et al., 2018).

A key aspect of the plastic pollution debate is understanding how plastic waste makes its way into the ocean. Much of this plastic travels through rivers before eventually reaching the ocean (Grossmann, 2023; Li et al., 2016). Globally, rivers are estimated to convey between 0.8 and 2.7 million metric tons of plastic into the oceans annually (Meijer et al., 2021), making rivers particularly relevant to consider as a source to tackle plastic pollution in the oceans.

Among the most significant contributing rivers is the Nile flowing through Egypt eventually into the Mediterranean Sea. The Nile delivers approximately 6772 tons (6.8%) of the plastic waste that enters the Mediterranean Sea each year (Liubartseva et al., 2018). Given Egypt's reliance on the Nile for water, transportation, and livelihoods, this pollution threatens both ecological balance and human well-being.

Research indicates that upper Egypt specifically experiences some of the highest levels of plastic waste accumulation, with areas such as Assiut, Nagaa Hammadi, and Sohag being particularly affected. These regions often receive limited attention from both governmental authorities and local communities, making them key areas of focus for targeted intervention and cleanup efforts (Hassan et al., 2024)

Thus, addressing plastic pollution particularly in the upper Nile River, is highly relevant and requires immediate and comprehensive action. With rising awareness and scientific evidence highlighting the scale of the problem, targeted strategies, improved waste management systems, and community-driven initiatives are necessary to mitigate the damage and protect the environment for future generations.

Building on the understanding of the problem's scale, this paper proposes a solution to the plastic pollution problem. It will then evaluate its potential benefits and limitations, and lastly present future visions and reflections.

2. Proposed Solutions

2.1 Floating Barriers

To address Egypt's role as a major contributor to plastic pollution in the Mediterranean, this paper proposes the deployment of floating barriers in key tributaries of the Nile River. Floating barriers offer a targeted and cost-effective solution that intercepts plastic waste before it reaches the sea, mitigating environmental damage while providing opportunities for community involvement and economic benefit. Such interventions, particularly in highly visible urban areas, can also raise public awareness

about plastic pollution and encourage greater environmental responsibility within local communities (Kumar et al., 2021).

Our team has chosen to install floating barriers in Assiut, located south of Egypt along the Nile River. As outlined in the problem section, Assiut represents a particularly critical area for intervention due to its high levels of plastic waste accumulation. The satellite image below highlights the region's distinctive river structure, where multiple branches and distributaries of the Nile create natural channels. These characteristics make Assiut especially suitable for the installation of floating barriers, as certain paths can be selectively blocked to intercept plastic waste while leaving other channels open for river traffic to flow unimpeded.

To ensure that navigation and local transportation are not disrupted, the floating barriers will be designed as modular and adjustable systems. This flexibility will allow for repositioning or partial openings as needed, accommodating changes in water flow, seasonal conditions, and traffic patterns. We believe that implementing floating barriers in Assiut holds strong potential for significantly reducing plastic pollution in the Nile River and, ultimately, preventing further contamination of the Mediterranean Sea.

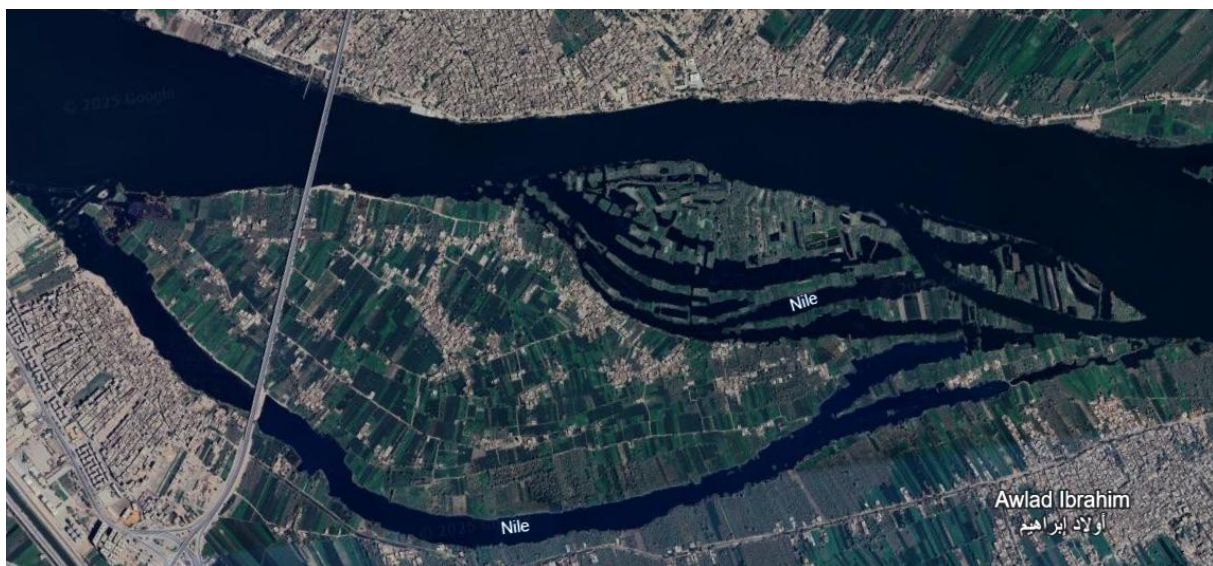


Figure 1: Nile at the Assiut governorate

Floating barriers have demonstrated considerable success in comparable contexts. For example, the *PROMAR Project* in the Caribbean has successfully piloted River Booms and Biobardas, showing their ability to trap both high and low volumes of floating plastic waste (Frank & Schindler, 2023).



Figure 2 & 3: Handmade Biobarda (Frank & Schindler, 2023, p. 22)

Biobardas, constructed from locally sourced and recycled materials, were installed in rivers in Colombia, achieving up to 90% reduction in floating solid waste in the Gaira and Manzanares Rivers (Frank & Schindler, 2023, p. 22). River Booms, as piloted in the Ozama River in the Dominican Republic, include mechanized systems powered by solar panels, helping scale the waste collection effort efficiently (Frank & Schindler, 2023, p. 6).



Figure 4: River Boom (Frank & Schindler, 2023, p. 10)

Similarly, the Ocean Cleanup's Interceptor is designed to extract plastic waste from rivers and has the capacity to collect up to 50 cubic meters of debris before needing to be emptied. In polluted rivers like the Rio Las Vacas in Guatemala, it has demonstrated

the ability to remove substantial amounts of plastic waste over time, significantly reducing the flow of debris into the ocean (The Ocean Cleanup, 2025).



Figure 5: Interceptor Barricade, Rio Las Vacas, Guatemala (The Ocean Cleanup)

While floating barriers are effective in intercepting plastic waste, their deployment requires careful planning to avoid disrupting local hydrology and harming marine habitats (European Environment Agency, n.d.). To minimize these risks, they will be installed in waterways with moderate currents and stable flow conditions, ensuring minimal impact on river dynamics. Comprehensive hydrological and environmental assessments will guide site selection, considering factors such as water velocity, seasonal floods, sediment transport, and proximity to plastic hotspots. To support this process and access proven technologies, we aim to collaborate with NGOs like *The Ocean Cleanup*. Once installed, the barriers will rely on the river's natural flow to guide floating debris toward a centralized collection point.

The project will follow a phased approach. Initially, Oceastic will deploy passive floating barriers that require no external energy, similar to the low-tech Biobardas used in Colombia (see above). These barriers can be constructed from recycled materials, such as plastic bottles and nets, and are ideal for low-flow tributaries. Once the system is established and community management structures are in place, the project will transition potentially to more advanced barriers. These will incorporate conveyor belts

and mechanized waste collection, powered by solar energy, as inspired by the River Boom as previously introduced. While automation reduces the need for labor-intensive collection in the water, it shifts labor toward activities like sorting, processing, and maintaining the system.

A central component of Oceastic is active community engagement, recognizing that long-term success depends on local management and participation. Drawing inspiration from successful initiatives like the *VeryNile*, Oceastic proposes the establishment of local community groups to oversee the operation and maintenance of the floating barriers. A collaboration with *VeryNile* would allow us to engage for example local fishermen and women in collecting plastic waste (VeryNile, n.d.).

Members will receive training in tasks such as the collection and sorting of plastic waste, as well as in the routine upkeep and monitoring of the barrier systems themselves. To further strengthen local capacity, educational workshops on environmental stewardship and waste reduction strategies will be integrated into the program. By embedding the initiative within local communities, Oceastic aims to promote a sustainable economy and ensure the barriers remain a lasting and effective solution.

2.2 Recycling Process and Product Development

After the collection and sorting of the plastic, we focus on transforming plastic waste into valuable products by categorizing it into two main types (UNIDO, 2024), each following a specific recycling track:

1- High-Density Polyethylene (HDPE) (Plastics for Change, 2023), found in plastic bottle caps and food containers, is ideal for creating strong, weather-resistant products. The process involves shredding the plastic into small pieces, melting it at around 200°C, and pressing it using hydraulic presses to eliminate air pockets. The plastic is then molded into chairs, tables, benches, and other long-lasting items, which are later trimmed and polished for durability and aesthetics. This process lowers production costs while producing sustainable, high-quality furniture. While melting has energy-related drawbacks, the benefits of the resulting products outweigh these disadvantages.



Figure 6: An example of recycled HDPE chair (CDR, 2023)

2- Low-Density Polyethylene (LDPE), found in single-use plastic bottles, is transformed into 3D printer filament, offering a cost-effective and eco-friendly alternative to traditional filament. The process begins with cleaning and drying the plastic, followed by shredding it into strands, melting, and extruding it through a nozzle to form a uniform filament. Once cooled and spooled, this filament becomes a raw material for 3D printing various products. We aim to collaborate with 3D printing agencies in Egypt such as FabMinds to supply them with affordable, recycled filament, reducing reliance on virgin plastics.



Figure 7: A visualization for the Filament with a machine (Recreator3D, 2023)



Figure 8: An example for the 3d printer potential printed Toy (Fab Minds,2023)

2.3 Funding Strategy

Our funding goal is to secure financial support from the government through the National Initiative for Smart Green Projects, operating under the Ministry of Planning and Economic Development (UN Egypt, 2024). This initiative focuses on strategic partnerships, including the United Nations Development Program (UNDP). In addition to government funding, we aim to establish a revenue stream by selling our recycled products. Moreover, the revenue will come from selling 3D printer filament as a cost-effective raw material to 3D printing agencies, offering competitive prices compared to existing filaments. These two approaches together will support our operations and enable us to provide financial incentives to our members, fostering economic empowerment within the local community.

3. Positive Impact

Oceastic is designed to intercept plastic waste and to generate positive environmental, economic, and social impacts. By engaging local communities, improving marine and river ecosystems, and promoting behavioral change, this solution aims to create long-term sustainability in Egypt's fight against plastic pollution.

By collecting plastic at an earlier stage, Oceastic supports SDG 14 (Life Below Water), and SDG 13 (Climate Action) by ensuring healthier aquatic environments and maintaining the ocean's role as a major CO₂ absorber. Additionally, SDG 6 (Clean Water and Sanitation) will be supported as polluted rivers are a health risk and harms

sanitation systems. By clearing out the rivers in such highly polluted areas, Oceastic can help to prevent this from happening.

Oceastic aligns with the concept of Bricolage, which involves using available resources innovatively. By repurposing collected plastic into new products, such as 3D-printed materials, Oceastic reduces waste while adding value to discarded plastics. Oceastic also helps with the issue of plastic consumption and production by limiting the production of new plastics through using the collected waste as raw materials for the new production, contributing to SDG 12 (Responsible Consumption and Production).

One of the most important aspects of Oceastic is its community-driven approach, which fosters local responsibility and long-term engagement. Instead of being a purely technological intervention, the initiative encourages collective action, bringing people together to solve the plastic problem in their own environment. This addresses SDG 11 (Sustainable Cities and Communities) by educating the public on plastic pollution and empowering them to take action. In addition to partnering with local communities, Oceastic partners with other entities and stakeholders like The Ocean Cleanup to use their experience and knowledge in developing our solution. Additionally, we plan to collaborate with the government to facilitate the installation of our solution, investors to fund us at the start of our solution and these various partnerships show Oceastic's support for SDG 17 (Partnerships For The Goals), as we create a network of valuable partnerships that we will try to expand globally in the future.

By embedding the project within Assiut's local communities, Oceastic avoids the "Broken Window Theory," which suggests that visible environmental neglect can contribute to further degradation and discourage community action. Instead, maintaining cleaner rivers and urban areas can promote a sense of pride and self-fulfillment, which encourages more sustainable behavior (Venhoeven et al., 2020).

Furthermore, Choice Architecture plays a key role in promoting behavioral change, in our case collecting plastic together with our initiative. The initiative will leverage social norm-driven decision-making by showing that community members are actively involved, fostering a feeling of participation in cleanup efforts being widespread and expected. Additionally, Conditional Cooperation can enhance participation, as people are more likely to contribute when they see others engaging in the same effort. By incorporating education and visual tools, such as tracking the amount of plastic

collected or highlighting before-and-after images of cleaned river sections, Oceastic will affirm and foster feelings of pride and accomplishment. This addresses the Intention-Action Gap and makes sustainable actions feel rewarding.

The Self-Determination Theory suggests that long-term participation in sustainability projects thrives when motivation shifts from external incentives (extrinsic motivation) to internal commitment (intrinsic motivation) (Dodds et al., 2020). In the future, Oceastic will provide financial and material incentives for participation, such as wages for waste collectors and product-based rewards for engagement. However, as awareness grows, the goal is for community members to feel intrinsically motivated to protect their environment, driven by a sense of responsibility and pride in having a cleaner city, rivers, and oceans.

4. Potential Downsides

While Oceastic presents a promising solution to mitigate plastic pollution, it is essential to recognize the potential downsides and challenges associated with its implementation. By repurposing plastic waste into new products using 3D printing technology, Oceastic may disrupt existing businesses that produce similar goods using traditional methods. This aligns with the concept of Creative Destruction, where innovation replaces older industries but may also create economic displacement for those whose livelihoods depend on conventional manufacturing. It also affects the people who used to provide plastics to companies for re-production. While the transition to sustainable production is necessary, it is essential to consider its economic impact on existing businesses and explore ways to integrate affected workers into the new system.

Due to the Anti-Authority Bias, where individuals resist information coming from perceived authority figures, particularly if they believe it undermines their autonomy or cultural values, efforts to educate local communities about plastic pollution might be met with resistance. Especially, if people perceive Oceastic as an external entity imposing knowledge upon them. To counteract this, Oceastic will focus on mutual learning, ensuring that education efforts feel collaborative and not imposed onto the people. This can also be supported through our group's constellation, as four of our members are Egyptians who could possibly help to bridge this gap.

Reactive Devaluation may also present a barrier if Oceastic is perceived as an external or elite-driven initiative. If local communities view the project as coming from

a different socio-economic or cultural "bubble", they may reject it, not because of the project itself, but because of who is proposing it. People may assume that outsiders do not understand their real priorities, leading to skepticism or non-participation.

The Self-Categorization Theory suggests that when people strongly identify with a group, they may perceive those outside the group as "the others." as it is also connected to the in-group-out-group effect (Turner & Reynolds, 2012). In this case, individuals actively participating in Oceastic might develop a sense of superiority over those who do not engage, potentially leading to division within the community. To prevent this, our initiative needs to create a feeling of inclusive participation, ensuring that involvement is accessible and welcoming to all.

Moreover, some individuals may respond with Defensive Denial when confronted with the emotional weight of environmental issues. If the messaging surrounding plastic pollution is too overwhelming or evokes feelings of helplessness or guilt, people may mentally disengage rather than act, prioritizing their own emotional well-being and needs over the problem at hand. Multidimensional poverty is high, affecting more than one in five individuals and MPI (Multidimensional Poverty Index)-poor people are on average deprived in more than one third of all indicators (Multidimensional poverty in Egypt, 2024, p.35). Results show that multidimensional poverty is unevenly distributed in Egypt. Rural areas record a higher MPI compared to urban parts of Egypt. Hence, we want to focus on empowering messaging to highlight our solutions as achievable.

Another challenge is that Oceastic focuses on one specific environmental issue, which could contribute to the Finite Pool of Worry effect. People have limited cognitive and emotional resources for concern, and by placing strong emphasis on plastic pollution, other critical issues such as poverty, healthcare, and unemployment may receive less attention. This is particularly significant in communities where environmental concerns may not be the most urgent priority (Sisco et al., 2020).

Another concern is the Single-Action Bias, which occurs when individuals take one positive action, such as cleaning plastic waste from a river, and feel that this alone is enough. As a result, they may be less likely to adopt other more impactful changes. If engagement in Oceastic leads people to believe they have "done their part" while continuing unsustainable behaviors in other areas of life, the long-term impact could be limited. The project must, therefore, ensure that participation is framed as part of an ongoing journey toward sustainability rather than a one-time contribution.

Additionally, the reliance on extrinsic motivation could undermine the Self-Determination Theory. If individuals engage in the initiative primarily due to potential incentives or tangible benefits, their participation may not be long-lasting. Once these incentives are removed or reduced, engagement levels may drop.

Finally, as mentioned in our proposed solution section, implementing floating barriers does come with risks like disruption of local hydrology and negative impacts on marine habitats and species. Here, working with mentioned NGOs like The Ocean Clean Up could help prevent this.

5. Future Vision and Reflection

We at Oceastic envision a future where communities, businesses, and governments are united in taking meaningful action to reduce waste and promote recycling. We have begun with **Phase 1 as an initial solution** and will continue to evolve and expand in future.

Phase 2: Expansion & Refinement (Months 7-12) Focus on implementation of Incentive-Based Plastic Collection Machines (IPCMs), development of an app, and launch of the initial product line of recycled plastic items.

We plan to implement IPCMs that allow individuals to deposit plastic waste and receive incentives based on its weight. Users will track their accumulated points via an app, which can be redeemed as vouchers at local businesses or used to purchase recycled products. The app will also feature geolocation to help users locate nearby machines. The collected plastic will be processed in the recycling stages, starting with Phase 1.

To boost community engagement, we will establish partnerships with, inter alia, women's cooperatives and orphanages, providing training and resources. We will also address hard-to-recycle materials like flip-flops, foams, and plastic bags through two main strategies. First, we will financially support local artisans and community members to repurpose these materials into handmade decorative accessories. Second, we will collaborate with industrial recycling partners, such as (TileGreen, n.d.), to efficiently process non-recyclable plastics into durable tiles for roads and buildings. This approach diversifies recycling methods, maximizes plastic waste utilization, and contributes to both economic value and environmental sustainability.

Additionally, we will develop a bilingual (Arabic & English) website to improve accessibility and outreach. The website will serve as a platform featuring updates,

impact stories, and a marketplace for recycled products. Social media content will be enhanced to showcase environmental benefits, community involvement, and product innovations. Small-scale marketing campaigns will be launched to raise local awareness and drive participation.



Figure 9: An example of handmade Flip Flop Art (Ocean Sole, 2023)

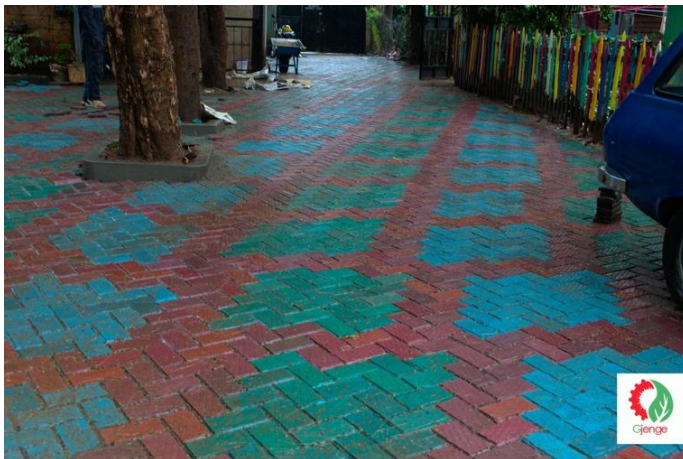


Figure 10: An example of Recycled Plastic Tiles (Gjenge Makers, 2023)

Phase 3: Scaling & Diversification (Months 13-18) Focus on scaling the incentive program, expanding recycled products, and strengthening partnerships.

Our focus also includes research and development (R&D) of 3D-printed plastic components for surveying drones, such as propeller guards and airframe parts. Drones will play a key role in mapping and tracking plastic waste, identifying plastic hotspots, and providing real-time data to optimize cleanup efforts. This technology will also help prevent waste buildup by monitoring areas before it becomes concentrated, supporting long-term solutions to reduce plastic pollution and improve waste management, in conjunction with floating barriers for better plastic collection in the ocean.

To strengthen the initiative, we will focus on building a stronger brand identity. We aim to form partnerships with local businesses and tourism operators to promote and sell recycled products, while also participating in regional markets and events to expand our reach. Additionally, we will explore collaborative projects, resource sharing, and advocacy opportunities by partnering with environmental NGOs focused on environmental education and river cleanup campaigns.

Phase 4: Sustainability & Impact (Months 19-24) Focus on ensuring the long-term sustainability of the initiative, maximizing environmental and social impact.

We will explore options for diversifying funding sources, such as corporate sponsorships and grants, to sustain and enhance the incentives. To expand the range and quality of recycled plastic products, we will invest in improved manufacturing equipment or processes, enabling us to create more diverse and higher-quality items.

If drone development proves successful, we will deploy the drones to monitor plastic waste distribution in the Nile, allowing us to assess the impact of the initiative over time. The data collected will be used to generate visualizations that support advocacy efforts and guide further actions. In addition, we will provide continued training and support to community members involved in recycling and product creation. Finally, our Oceastic prototype products ***“Where every piece tells a story of Nile”***

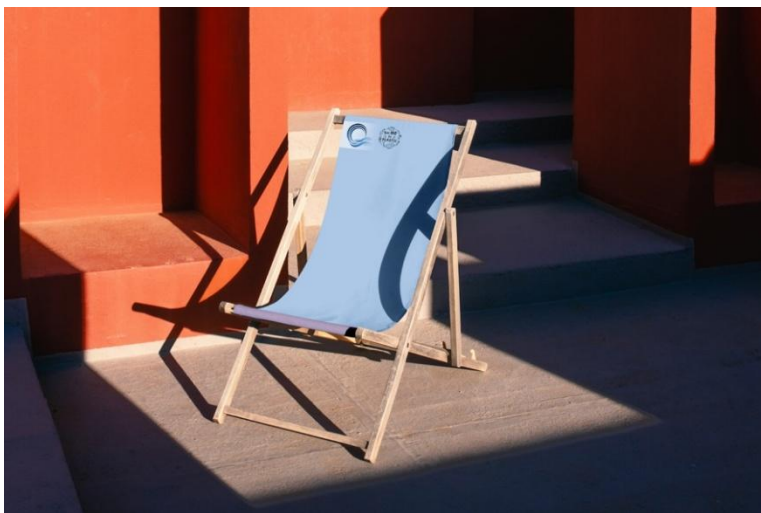


Figure 11: Prototype 1



Figure 12: Prototype 2



Figure 13: Prototype 3



Figure 14: Prototype 4

Coming back to the initial question: Are floating barriers the solution to plastic pollution in Egypt? Floating barriers are certainly not *the* solution, but they can be an important part of a broader, integrated strategy. When combined with community engagement, sustainable recycling practices, and partnerships across sectors, they offer a practical and scalable intervention to intercept plastic waste before it reaches the sea. While no single solution can address the complexity of plastic pollution, initiatives like Oceastic demonstrate how targeted action, innovation, and collaboration can create meaningful and lasting impact by protecting Egypt's waterways and contributing to a more sustainable future.

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